

SUPANTHO RAKSHIT

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EDUCATION

Princeton University

August 2021 - Present (*Ph.D. candidate*)

M.A. in Electrical and Computer Engineering 2023

Theoretical Quantum Optics Group.

Columbia University in the City of New York

August 2017 - May 2021

B.A. in Physics and Applied Mathematics

Science Research Fellow Class of 2021. Major GPA: 3.7/4.0

ACADEMIC INTERESTS AND TECHNICAL STRENGTHS

Emergent phenomenon in Large Language Models, Construction Grammar, Large Language Models, Information theory, Topological Quantum Error Correcting Codes

Programming languages: Python, Julia (*Advanced*), R, Qiskit, Mathematica (*Intermediate*)

Natural languages: Bengali, Hindi-Urdu, Persian, Russian, Hebrew, Tajik, Dari, English.

FELLOWSHIPS

Princeton AI Lab Graduate Student Fellowship — Natural and Artificial Minds (NAM)

AY 2025 – 26

- Organizing a reading group and lab sessions on Mechanistic interpretability, which is an emerging field that aims to explain the internal workings of AI models, providing insights into their decision-making processes.

WORK EXPERIENCE AND PROJECTS

Princeton Psychology of Language Lab

May 2024- present

Research Assistant

- Working on a project to study emergent structures of construction grammar in Large Language Models using tools from Mechanistic Interpretability and Information Theory

Tureci Group at Princeton ECE Department

September 2021- May 2024

Research Assistant

- Worked on a project utilizing a Quantum Reservoir for decoding error chains in degenerate quantum error correcting codes

Asenjo-Garcia Group at Columbia Physics Department

January 2020- May 2021

Research Intern

- Worked on a novel photonic implementation of measurement-based quantum computation. The theoretical proposal utilized a single ring resonator with two independent synthetic dimensions with a time-delayed feedback to achieve multidimensional cluster states.

Columbia Engineering and The Pierce Lab at Harvard

May- August 2018

Research Intern

- Developed opto-electronic tools and imaging techniques for experimentation on butterfly wing samples for analyzing the thermal and mechanical sensory properties of butterfly wings. (collaboration b/w Columbia School of Engineering and Harvard University).

TEACHING EXPERIENCES

Teaching Assistantships At Princeton ECE Department

Intelligent Robotic Systems (Spring 2024), Technology and Society (Spring 2024), Quantum Optics (Spring 2022), Foundations of Engineering: Mechanics, Energy, and Waves Mechanics (Fall 2022)

Teaching Assistantships At Columbia Physics Department

Accelerated Physics (Spring 2021) and Classical Mechanics (Fall 2020)

PUBLICATIONS AND CONFERENCES

Similarity All The Way Up: Multilingual Generalization in LLMs Relies on Language-Level Similarity Structures

CogSci 2026

Oral presentation & Diversity and Inclusion Award, CogSci 2026. Multilingual LLMs implicitly recover Indo-European phylogenetic structure in representation space, and the degree of alignment predicts cross-lingual (XNLI) transfer.

Meaning-infused grammar: Gradient Acceptability Shapes the Geometric Representations of Constructions in LLMs

August 2025

Proceedings of the 2nd International Workshop on Construction Grammars and NLP, ACL Anthology

For GPT-4 as with Humans: Information Structure Predicts Acceptability of Long-Distance Dependencies

July 2025

Proceedings of CogSci Conference 2025

A Practical Quantum Reservoir Computing Platform for Quantum Data Processing

March 2024

Bulletin of the American Physical Society

Reservoir Computing for Efficient Decoding of Error Syndromes in Surface Code

March 2024

Bulletin of the American Physical Society

On the Changes in Optical Interferometry Induced by the Relativistic Motion of an Optical Medium

September 2016

European Journal of Physics 37 (6), 065302

TALKS AND PRESENTATIONS

A Brief Exploration of Cluster State Quantum Computing

October 2020

Columbia Undergraduate Research Colloquium 2020

Mathematical Intricacies of Heisenberg's Matrix Mechanics Formulation of Quantum Mechanics

June 2020

Columbia Society of Physics Students Colloquium

Formulation of Schrodinger's Wave Mechanics from Hamilton-Jacobi Equation Using Classical Optics as a Guide

Nov 2019

Columbia Society of Physics Students Colloquium

Regulation and Optimization of Temperature of Butterfly Wings via Its Spatial Distribution of Coloration and Emissivity

August 2018

Undergraduate Science and Engineering Research Showcase 2018